

Base Ridge Learning Experiment

Question and design. This experiment asks whether the broker’s ranking advantage persists when both agents and the broker use Ridge regression instead of neural networks. Agents predict match quality from the candidate’s type. The broker uses additive and bilinear features of the pair and pools its mediated-match history. Thus, the experiment tests whether neural-network flexibility is necessary for a broker ranking advantage while retaining each learner’s information and feature map. The calibrated penalties are $\lambda_a = 0.001$ and $\lambda_b = 0.001$.

The primary estimand in realization c is

$$G_L(c) = \bar{r}_{b,L}(c) - \bar{r}_{a,L}(c),$$

where $L \in \{\text{NN}, \text{Ridge}\}$ and each bar is the seed-averaged holdout rank correlation over periods 481–500. The learning-system contrast is $G_{\text{Ridge}}(c) - G_{\text{NN}}(c)$. It includes the endogenous changes in matching and later learning histories induced by the learning rule. It is not a fixed-data comparison of estimators. Each of the 80 effective realizations receives one observation. Grid coordinates that refer to the same realization receive no additional weight. The model comparison uses the same seeds in both learning conditions: 50 at the baseline and 20 elsewhere.

Figures R1–R4 reproduce the corresponding main figures with NN and base Ridge on shared axes. Figure R5 shows paired comparisons of the primary ranking and output outcomes.

Baseline. With the common seeds, the NN broker-minus-principal difference in holdout rank correlation is 0.318, compared with 0.374 under base Ridge. The paired change is +0.056, with a 95% Monte Carlo interval of [+0.037, +0.074]. Table R1 reports other baseline outcomes. Figure R1 shows the full baseline dynamics, and Figure R5A isolates the broker and agent ranking trajectories.

Table R1: Baseline late-period outcomes under NN and base Ridge learning

Outcome	NN	Ridge	Ridge – NN
Agent holdout rank correlation	0.630	0.471	-0.160
Broker holdout rank correlation	0.948	0.845	-0.104
Broker-minus-principal difference in holdout rank correlation	0.318	0.374	+0.056
Brokered-minus-self output gap	1.713	1.564	-0.150
Outsourcing rate	0.968	0.942	-0.026
Broker access fraction	0.112	0.116	+0.005
Broker betweenness centrality	0.790	0.749	-0.041

Entries use the 50 common seeds. Each seed is first averaged over periods 481–500. The Ridge-minus-NN column is the mean paired seed difference. The access fraction is the share of accepted broker offers whose counterparty was not already the demander’s neighbor.

Robustness across model realizations. Across the 80 effective realizations, the median broker-minus-principal difference in holdout rank correlation is 0.271 for NN and 0.354 for Ridge. The median Ridge-minus-NN contrast is +0.069. Its 10th and 90th percentiles are -0.038 and +0.268. The Ridge difference is positive in 80 realizations, compared with 80 under NN, and exceeds the NN difference in 64 realizations. The correlation between NN and Ridge condition-level differences in holdout rank correlation is 0.277.

The within-system brokered-minus-self output gap has medians of 1.541 under NN and 1.340 under Ridge. Its median Ridge-minus-NN contrast is -0.201, and Ridge exceeds NN in 11 realizations. This channel mean difference is descriptive, not a causal estimate of the value of brokerage: channel choice, candidate sets, and offer acceptance are endogenous. Median outsourcing is 0.963 under NN and 0.934 under Ridge.

Figure R2 shows the matching-problem surface under each learning system. Figure R3 compares the broker’s network position and access. Figure R4 relates the ranking and output differences to those structural outcomes. Figure R5B–C shows the paired condition-level changes directly against zero.

Interpretation. Neural-network flexibility is not necessary for the broker to rank candidates better than agents in this modeled system. The Ridge broker-minus-principal difference is positive in 80 of 80 effective realizations. This experiment does not identify the source of that advantage because the broker still observes both parties, represents pairs with additive and bilinear features, and pools mediated-match observations. The ablation experiment restricts these features separately.

Illustrative figures

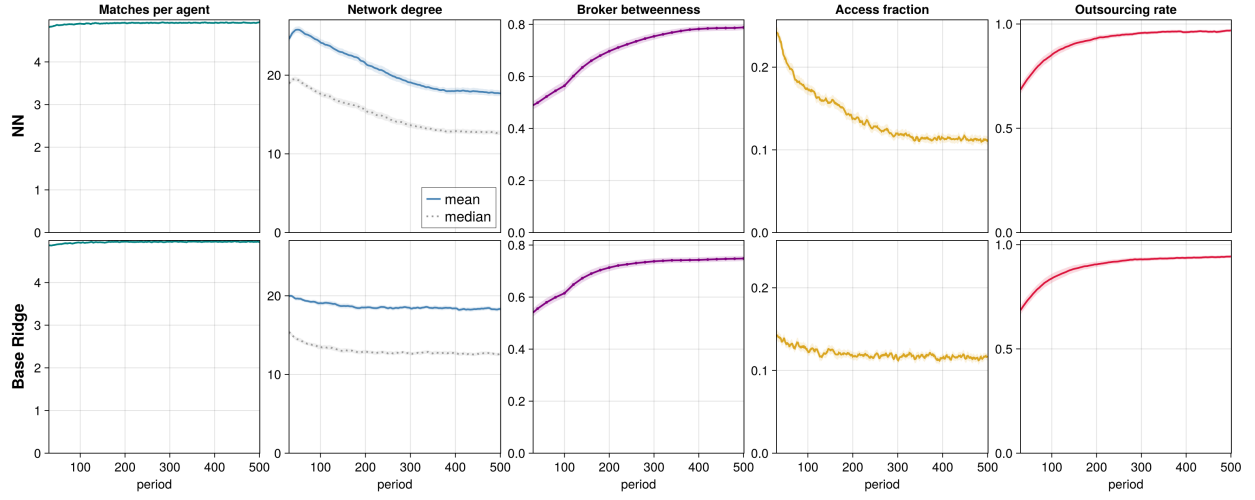


Figure R1: Baseline dynamics under NN and base Ridge. Rows identify the learning system. Panels show matches per agent, mean and median agent-network degree, broker betweenness centrality, access fraction, and outsourcing rate. Lines are 5-observation rolling ensemble means over 50 common seeds. Ribbons are pointwise 95% Monte Carlo intervals. Betweenness is measured every 20 periods; other measures are recorded every period. Axes begin at period 30 and are shared across learning systems.

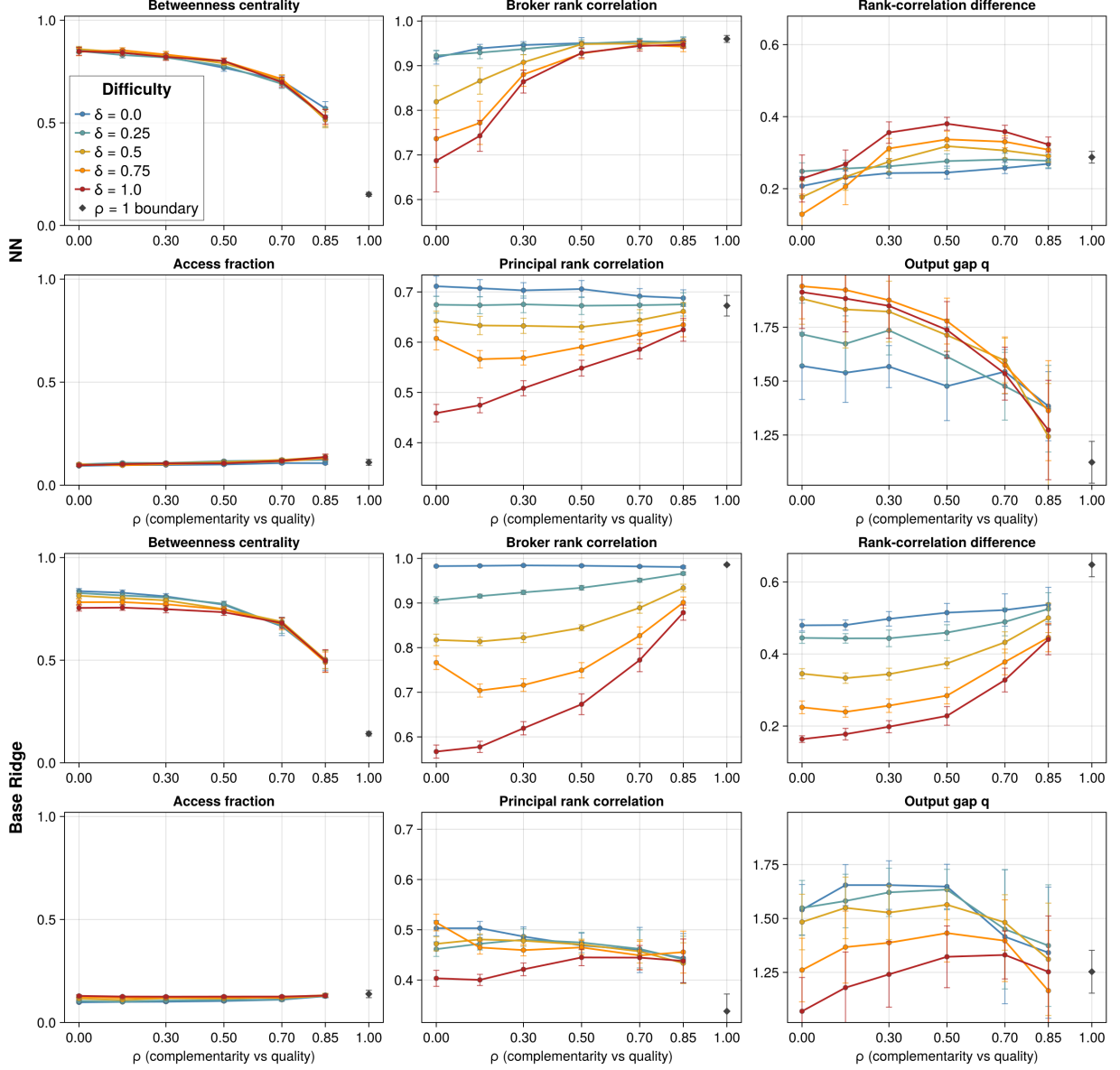


Figure R2: Outcomes across the $\rho \times \delta$ matching grid under NN (top two rows) and base Ridge (bottom two rows). Each line is a value of δ ; lines span $\rho = 0.0$ through 0.85. The pure-quality boundary $\rho = 1.0$, where δ is inactive, is shown once and unconnected. Each marker is an effective realization's late-window seed mean. Corresponding NN and Ridge panels use the same vertical scale. Whiskers are 95% Monte Carlo intervals.

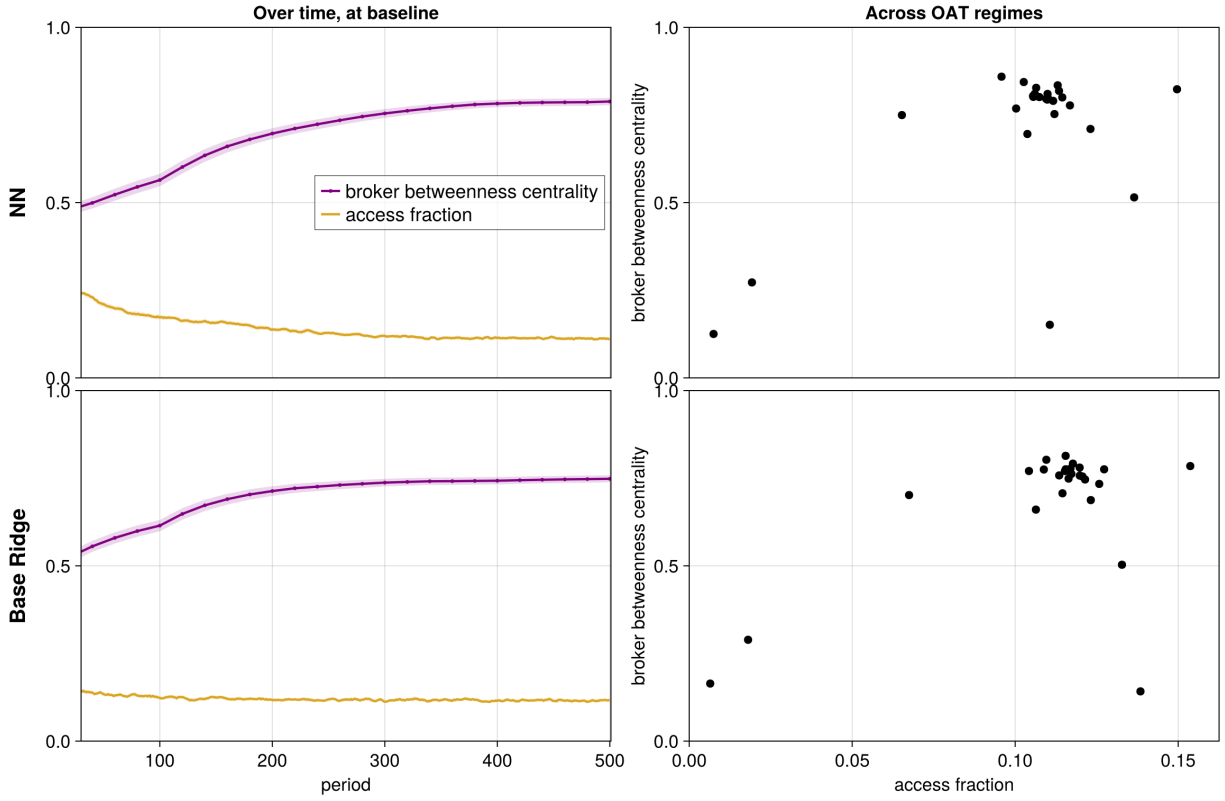


Figure R3: Broker position and access under NN and base Ridge. Left panels show 5-observation rolling ensemble means at the baseline. Right panels show one point per one-at-a-time regime, using late-window seed means. Ribbons are pointwise 95% Monte Carlo intervals. Corresponding panels use shared axes.

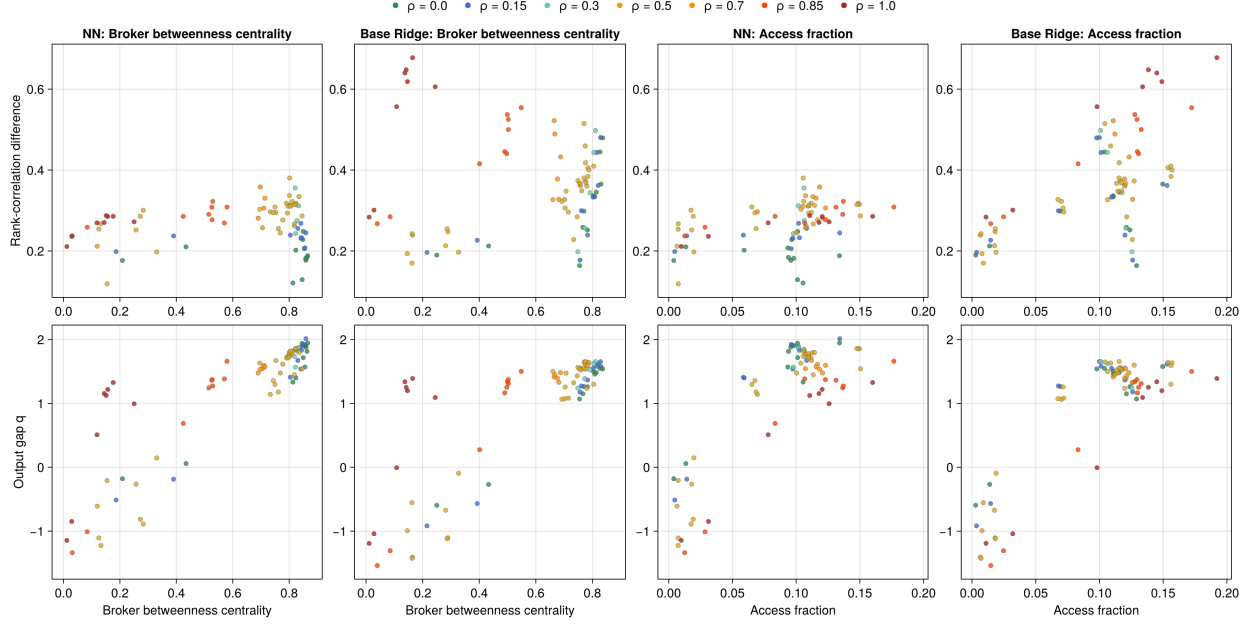


Figure R4: Broker-minus-principal differences in holdout rank correlation and brokered-minus-self output gaps against broker betweenness centrality and access fraction under NN and base Ridge. Each point is one effective realization's late-window seed mean. Corresponding NN and Ridge panels use shared axes; color denotes ρ .

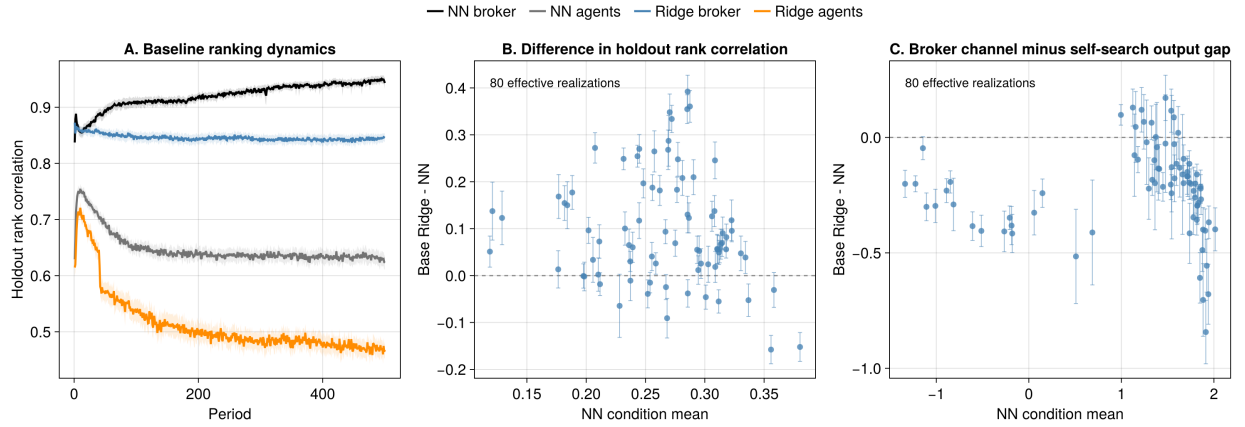


Figure R5: Direct comparison of NN and base Ridge. Panel A shows broker and principal holdout rank correlation over time under NN and base Ridge at the baseline. Lines are ensemble means over 50 common seeds; ribbons are pointwise 95% Monte Carlo intervals. Panels B and C show the paired change from NN to base Ridge in the broker-minus-principal difference in holdout rank correlation and in the broker channel-minus-self-search output gap, respectively, for each of the 80 effective realizations. The horizontal position is that realization's NN mean. Whiskers are paired 95% Monte Carlo intervals for the mean seed-level change, and dashed lines mark no change.